Niraj Kumar

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Professional Summary

Senior Data Scientist with 4+ years building production ML and GenAI systems across fraud detection and ride-hailing marketplaces. Owns the full lifecycle, from problem framing and experiment design to model building, LLM fine-tuning, and deployment. Delivered $1M+ in impact by automating ∼60% of a manual fraud review process.

Professional Experience

VinFast (GSM - Green SM) Aug 2026 – Present

Data Scientist II

- Building a **dynamic dispatch system that adjusts matching parameters (search radius, pickup distance, batching window) in real time based on local supply and demand**
- Analyzing how these parameters affect acceptance-rate, ETA, match rate, and cancellations, to replace static city-level settings with adaptive, data-driven ones

Accertify (Demerged from American Express) - AI Team May 2024 – Jul 2026

Senior Data Scientist

- Designed **ML decisioning systems** using gradient boosted trees and rule engines, **auto-resolving ∼60% of transaction review queues and delivering $1M+ in operational cost savings and fraud-dollar recovery**
- **Engineered a 100M+ record, 250+ feature PySpark dataset** augmented with external risk signals; established automated daily monitoring dashboards using Prometheus and Grafana
- **Conducted A/B testing and canary release stress tests to optimize decisioning thresholds**, with automated daily performance tracking across critical client segments
- **Fine-tuned LLaMA 3, Mistral 7B, and Qwen for fraud classification using QLoRA-based PEFT via HuggingFace Transformers to capture sequential context between transactions; outperformed the legacy XGBoost baseline for high-imbalance fraud classification**
- **Collaborated with Product, Engineering, and Client Success** to align model outputs to ROI-driven KPIs; **mentored junior data scientists** and standardized reproducibility and monitoring practices

American Express - Global Decision Science Aug 2022 – Apr 2024

Data Scientist

- **Awarded Employee of the Quarter** for building a first-of-its-kind fraud detection framework for the Money Transfer vertical, cutting false negative rates and chargeback exposure
- **Built gradient boosted fraud detection models across Money Transfer, Retail, and Travel verticals; deployed via FastAPI microservices on AWS**
- Designed ML-layered rule engines to handle seasonal fraud spikes; **developed a label denoising method to correct over-marked training samples, reducing target variable noise by ∼15%**
- **Automated daily model monitoring using Airflow**, reducing analyst reporting effort

Education

Indian Statistical Institute Jul 2020 – Jul 2022

M.Tech in Computer Science *Kolkata, India*

Thesis: Reconstruction and Prediction based approaches for Anomaly Detection in Surveillance Videos

Technical Skills

Data Manipulation and Programming Languages: C, Python, SQL, Bash, Pandas, NumPy, PyTorch
Machine Learning: Scikit-learn, Gradient Boosted Trees, Regression, Classification, Clustering, Anomaly Detection
Generative AI and LLMs: Transformers, PEFT (LoRA, QLoRA), LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Vector DBs, Supervised Fine-Tuning (SFT), RAGAS
MLOps and Infrastructure: MLflow, Airflow, Docker, FastAPI, CI/CD, Prometheus, Grafana, AWS, Git, Apache Spark (PySpark)
Core Competencies: Risk and Marketplace Analytics (Fraud Detection, Dispatch Optimization), Gen-AI and Engineering (Large Language Models (LLMs), Model Deployment, Monitoring), Natural Language Processing (NLP)